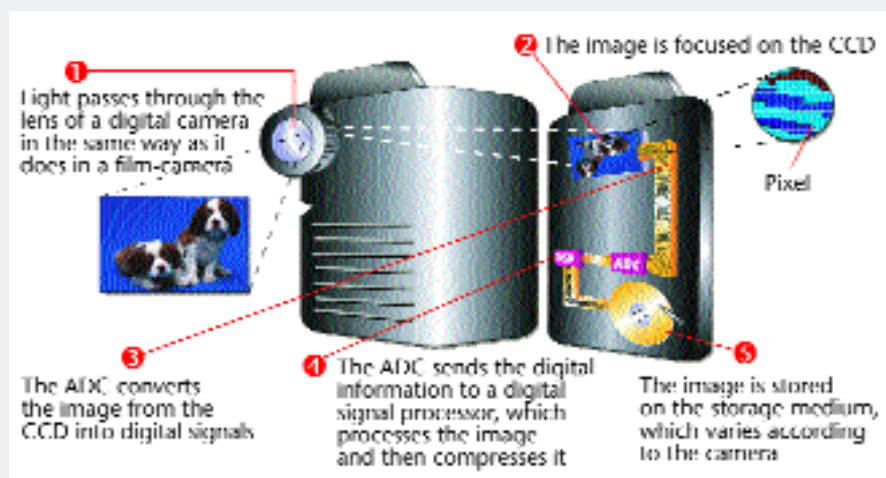


How it Works

A digital camera works in the same manner as a conventional camera as far as optics are concerned, with one main difference—conventional cameras use film to sense and store the image while digital cameras sense the image on a CMOS or CCD element, which is a light-sensitive element that consists of millions of pixels. The CCD is an array of very tiny photo transistors that are arranged in a grid. When this grid is exposed to light, it generates electricity depending upon the intensity of light falling on it. These transistors are the smallest elements on the sensing device and they form the pixels in the image. Therefore, the greater the number of pixels on the CCD, greater will be the resolution supported and consequentially, greater is the clarity. On a monochrome digital camera, one pixel would be made up of a single transistor but in a colour digital camera, each pixel would be composed of three separate transistors, red, green and blue. Together these three colours can represent any number of colours depending upon the light falling on them.



These transistors generate continuous electrical signals that are sent to an Analog to Digital Converter (ADC) where these signals are converted to a digital format. This is then sent to a specialised processor called a DSP (Digital Signal Processor) that is programmed specifically for handling image information, where settings such as white balance, filters and image compression are performed. Finally, the image is sent to the storage memory.

Another very integral part of most digital cameras is the LCD screen that lets the user view the settings the camera is using, edit, organise and preview pictures as well as access and change the mode of its operation.

These digital images can now be viewed on an external display device such as a monitor or a television and can be downloaded to the computer using the TWAIN driver provided by the manufacturer.

How they fared

We found a wide variation in the implementation of features and performance in the 17 cameras tested. Each price category had its own strengths—the sub-2.0 megapixel cameras provided bare bones functionality, while the 2+ megapixel cameras offered a good balance between features and performance.

Features

At their functional core, digital cameras differ only in their CCDs and the optics used. Therefore, the primary factor that sets apart one from the other in the same category is the number of feature it includes. This is especially useful for semi-professional users who would want the maximum amount of storage possible, plenty of imaging controls, and flash and focusing modes to experiment with. Amateur users would also want features such as the ability to use the camera as a Web cam and to capture video clips.

White balance: In the sub-2.0 megapixel category, the camera with the maximum number of white balance settings was Olympus Camedia C-1 and Fuji FinePix A101. Each had seven white balance settings including clear sky, incandescent lamp, fluorescent as well as an

Auto setting. Though the cameras in this category did not have the ability to manually calibrate the white balance, the presets available on them introduced a very intuitive way for implementing white balance, involving minimal user intervention—something that novices at photography will appreciate.

The lower-end cameras such as the D-Link DSC-350 and the Benq DC300 did not have any means of selecting the white balance settings. Since these cameras were not capable of capturing images above a resolution of 640x480, you cannot expect them to sport advanced features like white balance and metering modes.

In the high-end category, the Fuji FinePix series had five white balance settings built into all three models, the FinePix 2800, FinePix 2600 and the FinePix A201. The camera that stood out from the pack here was the Nikon CoolPix 800, which had five presets, an auto white balance mode and also manual settings for the white balance level—invaluable to photography enthusiasts. No other camera in this roundup supported manual calibration of white balance.

Resolution: All the cameras in the sub-2.0 megapixel range (save the Benq DC300 and D-Link DDSC-350) sported a

native resolution of 1280x960. At this resolution you can take 4.26x3.2 inch prints at 300 dpi. The truly outstanding camera here was the Casio GV-10, which could capture images at a resolution of 1600x1200. This camera uses a Pixel Generator that effectively 'fills in' the pixels if you were to use the images for printing A4-size prints.

On the lower end of the scale, the D-Link DSC-350 and the Benq DC300 could only capture images at a resolution of 640x480 and this was attributed to the very low-density CCDs which gave them Web cam-class imaging capabilities. In fact, they just about enter the realm of digital cameras because of their small form factor, integrated memory and portability. All the 1.3 megapixel cameras in this category are suitable mostly for Web-based applications. If you need to print photographs, you would have to lower the resolution and settle for really small images; if you scale the images to fit the industry standard of 4x6 inch or higher prints, it would result in a loss in quality. The only camera that allowed such print resolutions was the Casio GV-10, though it did this through image processing.

In the 2+ megapixel category, the resolution across the cameras was primarily 1600x1200, with the exception of the